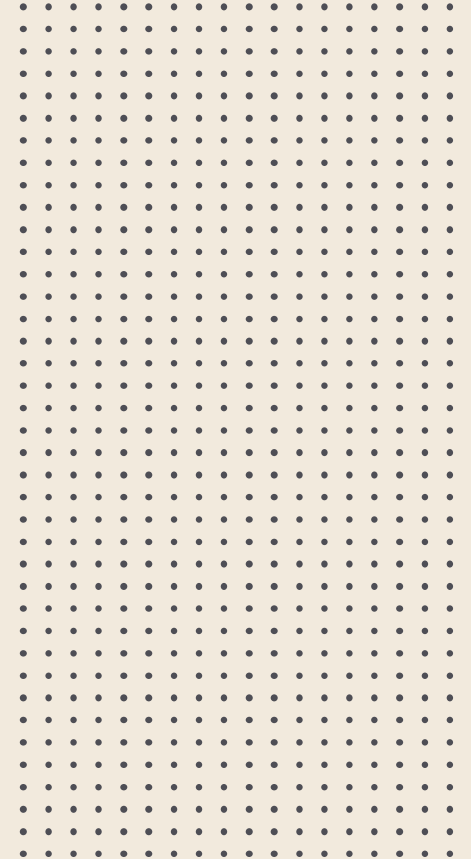


CSE-1223

Discrete Mathematics

Segment 5: Mathematical Reasoning

Md. Saiful Islam
Lecturer, Dept. of CSE, IIUC





Methods of Proof

- ❑ A **proof** is a valid argument that establishes the truth of a mathematical statement
- ❑ **Axiom** (or **postulate**): a statement that is assumed to be true
- ❑ **Theorem**
A statement that has been proven to be true
- ❑ **Hypothesis, premise**
An assumption (often unproven) defining the structures about which we are reasoning



Methods of Proof

□ **Lemma**

A minor theorem used as a stepping-stone to proving a major theorem.

□ **Corollary**

A minor theorem proved as an easy consequence of a major theorem.

□ **Conjecture**

A statement whose truth value has not been proven. (A conjecture may be widely believed to be true, regardless.)



Methods of Proof

Informal proofs: The proofs of theorems designed for human consumption are almost always informal proofs, where more than one rule of inference may be used in each step, where steps may be skipped, where the axioms being assumed and the rules of inference used are not explicitly stated. **Informal proofs** can often explain to humans why theorems are true, while computers are perfectly happy producing formal proofs using automated reasoning systems.

Direct Proof: A direct proof of a conditional statement $p \rightarrow q$ is constructed when the first step is the assumption that p is true; subsequent steps are constructed using rules of inference, with the final step showing that q must also be true. [**Assume p is true, and prove q**]



Methods of Proof

Indirect Proof: Instead of proving $p \Rightarrow q$ directly, it is sometimes easier to prove it indirectly. There are two kinds of **indirect proofs**: the proof by contrapositive, and the proof by contradiction.

The **proof by contrapositive** is based on the fact that an implication is equivalent to its contrapositive. Therefore, instead of proving $p \Rightarrow q$, we may prove its contrapositive $\neg q \Rightarrow \neg p$.

Another indirect proof is the **proof by contradiction**. To prove that $p \Rightarrow q$, we proceed as follows: Suppose $p \Rightarrow q$ is false; that is, assume that p is true and q is false.



Methods of Proof

Example 1: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.” [Spr’18]

Solution: Note that this theorem states $\forall n P(n) \rightarrow Q(n)$, where $P(n)$ is “ n is an odd integer” and $Q(n)$ is “ n^2 is odd.” As we have said, we will follow the usual convention in mathematical proofs by showing that $P(n)$ implies $Q(n)$, and not explicitly using universal instantiation. To begin a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that n is odd. By the definition of an odd integer, it follows that $n = 2k + 1$, where k is some integer. We want to show that n^2 is also odd. We can square both sides of the equation $n = 2k + 1$ to obtain a new equation that expresses n^2 . When we do this, we find that $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. By the definition of an odd integer, we can conclude that n^2 is an odd integer (it is one more than twice an integer). Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer. ◀

Methods of Proof

Example 2: Give a direct proof that if m and n are both perfect squares, then mn is also a perfect square. (An integer a is a **perfect square** if there is an integer b such that $a = b^2$.)

Solution: To produce a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that m and n are both perfect squares. By the definition of a perfect square, it follows that there are integers s and t such that $m = s^2$ and $n = t^2$. The goal of the proof is to show that mn must also be a perfect square when m and n are; looking ahead we see how we can show this by substituting s^2 for m and t^2 for n into mn . This tells us that $mn = s^2 t^2$. Hence, $mn = s^2 t^2 = (ss)(tt) = (st)(st) = (st)^2$, using commutativity and associativity of multiplication. By the definition of perfect square, it follows that mn is also a perfect square, because it is the square of st , which is an integer. We have proved that if m and n are both perfect squares, then mn is also a perfect square. 



Methods of Proof

Example 3: Give a direct proof of the theorem “If n is an odd integer, then show that n^2-1 is divisible by 8”. [Aut’23]

Solution:

We know that,

Odd number is in the form of $(4k+1)$ and $(4k+3)$ where k is a natural number,

If n is an odd integer, then show that n^2-1 is divisible by 8 When $n = (4k+1)$

$$\text{so, } n^2-1 = (4k+1)^2-1$$

$$= 16k^2 + 8k + 1 - 1$$

$$= 16k^2 + 8k$$

$$= 8(2k^2 + k) \text{ is divisible by 8.}$$

If n is an odd integer, then show that n^2-1 is divisible by 8 When $n = (4k+3)$

$$\text{so, } n^2-1 = (4k+3)^2-1$$

$$= 16k^2 + 24k + 9 - 1$$

$$= 16k^2 + 24k + 8$$

$$= 8(2k^2 + 3k + 1) \text{ is divisible by 8.}$$

So, we can say that If n is an odd integer, then n^2-1 is divisible by 8.



Methods of Proof

Example 4: Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.
[Spr'17]

Solution: Suppose, on the contrary, $\sqrt{2}$ is rational. Then we can write

$$\sqrt{2} = \frac{m}{n}$$

for some positive integers m and n such that m and n do not share any common divisor except 1 (hence mn is in its simplest term). Squaring both sides and cross-multiplying yields

$$2n^2 = m^2$$

Thus, 2 divides m^2 . Consequently, 2 must also divide m . Then we can write $m = 2s$ for some integer s . The equation above becomes

$$2n^2 = m^2 = (2s)^2 = 4s^2$$

$$\text{Hence, } n^2 = 2s^2$$

which implies that 2 divides n^2 ; thus, 2 also divides n . We have proved that both m and n are divisible by 2. This contradicts the assumption that m and n do not share any common divisor. Therefore, $\sqrt{2}$ must be irrational.



Methods of Proof

Exercise:

1. Let n be an integer. Prove that if n^2 is even, then n must be even. Use
 - A proof by contrapositive.
 - A proof by contradiction.
2. Prove that $\sqrt{3}$ is irrational.
3. Let n be an integer. Show that if n^2 is even, then n is also even.



Mathematical Induction

Mathematical Induction is a technique of proving a statement, theorem or formula which is thought to be true, for each and every natural number n . By generalizing this in form of a principle which we would use to prove any mathematical statement is '**Principle of Mathematical Induction**'.

For example: $1^3 + 2^3 + 3^3 + \dots + n^3 = (n(n+1)/2)^2$, the statement is considered here as true for all the values of natural numbers.



Mathematical Induction



Principle of Mathematical Induction: [Aut'16, Spr'16, Aut'22]

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .



Mathematical Induction



Why Mathematical Induction is Valid? [Spr'16, Aut'17, Aut'22]

Why is mathematical induction a valid proof technique? The reason comes from the well-ordering property, listed in Appendix 1, as an axiom for the set of positive integers, which states that every nonempty subset of the set of positive integers has a least element. So, suppose we know that $P(1)$ is true and that the proposition $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . To show that $P(n)$ must be true for all positive integers n , assume that there is at least one positive integer for which $P(n)$ is false. Then the set S of positive integers for which $P(n)$ is false is nonempty. Thus, by the well-ordering property, S has a least element, which will be denoted by m . We know that m cannot be 1, because $P(1)$ is true. Because m is positive and greater than 1, $m - 1$ is a positive integer. Furthermore, because $m - 1$ is less than m , it is not in S , so $P(m - 1)$ must be true. Because the conditional statement $P(m - 1) \rightarrow P(m)$ is also true, it must be the case that $P(m)$ is true. This contradicts the choice of m . Hence, $P(n)$ must be true for every positive integer n .

Mathematical Induction

Example 1: Use mathematical Induction to show that if n is a positive integer, then $1 + 2 + \cdots + n = n(n + 1)/2$.

Solution:

Let,

$$P(n) : 1+2+3+\dots+n = \frac{n(n+1)}{2}$$

Basis Step: We must verify that $P(1)$ is true.

For $n=1$,

$$\text{L.H.S.} = 1$$

$$\text{R.H.S.} = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(n)$ is true for $n=1$.

Inductive Step: Assume $P(k)$ to be true and then prove

$P(k+1)$ is true.

Assume that $P(k)$ is true,

$$P(k) : 1+2+3+\dots+k = \frac{k(k+1)}{2} \text{ ----- (1)}$$

We will prove that $P(k+1)$ is true.

$$P(k+1) : 1+2+3+\dots+(k+1) = \frac{(k+1)(k+1+1)}{2}$$

$$P(k+1) : 1+2+3+\dots+k+(k+1) = \frac{(k+1)(k+2)}{2}$$

$$\begin{aligned} \text{L.H.S.} &= 1+2+3+\dots+k+(k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \quad [\text{From equation 1}] \\ &= \frac{k(k+1)+2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \\ &= \text{R.H.S.} \end{aligned}$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(k+1)$ is true when $P(k)$ is true.

Mathematical Induction

Example 2: Use mathematical induction to prove that the sum of the n first odd positive integers is n^2 , i.e., $1 + 3 + 5 + \cdots + (2n - 1) = n^2$. [Spr'16, Aut'16,'22]

Solution :

Let,

$$P(n) : 1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Basis Step: We must verify that $P(1)$ is true.

For $n=1$,

$$\text{L.H.S.} = 2 \cdot 1 - 1 = 2 - 1 = 1$$

$$\text{R.H.S.} = 1^2 = 1$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(n)$ is true for $n=1$.

Inductive Step: Assume $P(k)$ to be true and then prove

$P(k+1)$ is true.

Assume that $P(k)$ is true,

$$P(k) : 1 + 3 + 5 + \cdots + (2k - 1) = k^2 \quad \text{----- (1)}$$

We will prove that $P(k+1)$ is true.

$$P(k+1) : 1 + 3 + 5 + \cdots + \{2(k+1) - 1\} = (k+1)^2$$

$$P(k+1) : 1 + 3 + 5 + \cdots + (2k+2 - 1) = (k+1)^2$$

$$P(k+1) : 1 + 3 + 5 + \cdots + (2k+1) = (k+1)^2$$

$$P(k+1) : 1 + 3 + 5 + \cdots + (2k - 1) + (2k+1) = (k+1)^2$$

$$\text{L.H.S.} = 1 + 3 + 5 + \cdots + (2k - 1) + (2k+1)$$

$$= k^2 + (2k+1) \quad [\text{From equation 1}]$$

$$= k^2 + 2k + 1$$

$$= (k+1)^2$$

$$= \text{R.H.S.}$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(k+1)$ is true when $P(k)$ is true.



Mathematical Induction

Example 3: Use mathematical induction to show that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n .

Solution:

Let,

$$P(n) : 1 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

Basis Step: We must verify that $P(1)$ is true.

For $n=1$,

$$\text{L.H.S.} = 1 + 2^1 = 3$$

$$\text{R.H.S.} = 2^{1+1} - 1 = 2^2 - 1 = 3$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(n)$ is true for $n=1$.

Inductive Step: Assume $P(k)$ to be true and then prove

$P(k+1)$ is true.

Assume that $P(k)$ is true,

$$P(k) : 1 + 2^1 + 2^2 + \cdots + 2^k = 2^{k+1} - 1 \quad \text{----- (1)}$$

We will prove that $P(k+1)$ is true.

$$P(k+1) : 1 + 2^1 + 2^2 + \cdots + 2^{k+1} = 2^{k+1+1} - 1$$

$$P(k+1) : 1 + 2^1 + 2^2 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1$$

$$\begin{aligned} \text{L.H.S.} &= 1 + 2^1 + 2^2 + \cdots + 2^k + 2^{k+1} \\ &= 2^{k+1} - 1 + 2^{k+1} \quad [\text{From equation 1}] \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{1+k+1} - 1 \\ &= 2^{k+2} - 1 \\ &= \text{R.H.S.} \end{aligned}$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(k+1)$ is true when $P(k)$ is true.

Mathematical Induction

Example 4: Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$ whenever n is a positive integer. [Aut'23]

Solution:

Let,

$$P(n) : 1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$$

Basis Step: We must verify that $P(1)$ is true.

For $n=1$,

$$\text{L.H.S.} = 1 \cdot 1! = 1 \cdot 1 = 1$$

$$\text{R.H.S.} = (1 + 1)! - 1 = 2! - 1 = 2 - 1 = 1$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(n)$ is true for $n=1$.

Inductive Step: Assume $P(k)$ to be true and then prove $P(k+1)$ is true.

Assume that $P(k)$ is true,

$$P(k) : 1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! = (k + 1)! - 1 \quad \text{-----} \\ (1)$$

We will prove that $P(k+1)$ is true.

$$P(k+1) : 1 \cdot 1! + 2 \cdot 2! + \cdots + (k+1) \cdot (k+1)! = (k+1 + 1)! - 1$$

$$P(k+1) : 1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k+1) \cdot (k+1)! = (k+2)! - 1$$

$$\begin{aligned} \text{L.H.S.} &= 1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k+1) \cdot (k+1)! \\ &= (k + 1)! - 1 + (k+1) \cdot (k+1)! \quad [\text{From equation 1}] \\ &= (k+1) \cdot (k+1)! + (k + 1)! - 1 \\ &= (k+1)! (k+1+1) - 1 \\ &= (k+1)! (k+2) - 1 \\ &= (k+2) (k+1)! - 1 \\ &= (k+2)! - 1 \\ &= \text{R.H.S.} \end{aligned}$$

Since, $\text{L.H.S.} = \text{R.H.S.}$

$\therefore P(k+1)$ is true when $P(k)$ is true.

Mathematical Induction

Example 5: Let $P(n)$ be the statement that $1^3+2^3+\dots+n^3 = (\frac{n(n+1)}{2})^2$ for the positive integer n . [Sp'23]

- a) What is the statement $P(1)$?
- b) Show that $P(1)$ is true, completing the basis step of the proof.
- c) What is the inductive hypothesis?
- d) What do you need to prove in the inductive step?
- e) Complete the inductive step, identifying where you use the inductive hypothesis.
- f) Explain why these steps show that this formula is true whenever n is a positive integer.

Answer :

a) For $n=1$, $P(1)$: $1^3 = (\frac{1(1+1)}{2})^2$.

b) L.H.S. $= 1^3 = 1$

$$\text{R.H.S.} = (\frac{1(1+1)}{2})^2 = (\frac{1*2}{2})^2 = 1^2 = 1$$

Since, L.H.S. $=$ R.H.S. , $\therefore P(n)$ is true for $n=1$. This completes the basis step.

c) For the inductive hypothesis, we assume that $P(k)$ is true for an arbitrary positive integer k . That is, we assume that

$$1^3+2^3+\dots+k^3 = (\frac{k(k+1)}{2})^2 \text{ ----- (1)}$$

Mathematical Induction

d) To carry out the inductive step we must show that when we assume that $P(k)$ is true, then $P(k+1)$ is also true. That is, we must show that,

$$1^3+2^3+\dots+k^3 + (k+1)^3 = \left(\frac{(k+1)(k+1+1)}{2}\right)^2 = \left(\frac{(k+1)(k+2)}{2}\right)^2. \text{ Assuming the inductive hypothesis } P(k).$$

$$\text{e) } P(k+1): 1^3+2^3+\dots+k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2$$

$$\text{L.H.S.: } 1^3+2^3+\dots+k^3 + (k+1)^3$$

$$= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \quad [\text{From equation 1}]$$

$$= \frac{k^2(k+1)^2}{4} + (k+1)(k+1)^2 = \frac{k^2(k+1)^2 + 4(k+1)(k+1)^2}{4} = \frac{(k+1)^2(k^2 + 4k+4)}{4} = \frac{(k+1)^2(k+2)^2}{4}$$

$$= \left(\frac{(k+1)(k+2)}{2}\right)^2$$

$$= \text{R.H.S.}$$

Since, L.H.S. = R.H.S. $\therefore P(k+1)$ is true when $P(k)$ is true.

We have completed the inductive step.

f) Because we have completed the basis step and the inductive step, by mathematical induction we know that $P(n)$ is true for all positive integers n . That is, $1^3+2^3+\dots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for the positive integer n .



Mathematical Induction

Example 6: Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer. [Aut'17, '20, '22]

Solution:

BASIS STEP: For $n=1$, we have $1^3 - 1 = 0$ is divisible by 3.

INDUCTIVE STEP: Assume that this is true for n and we have to show that it's true for $n+1$.

$$\begin{aligned}\therefore (n+1)^3 - (n+1) &= n^3 + 3n^2 + 3n + 1 - n - 1 \\ &= n^3 - n + 3n^2 + 3n \\ &= (n^3 - n) + 3(n^2 + n) \text{ is divisible by 3;}\end{aligned}$$

$n^3 - n$ is divisible by 3 by assumption and $3(n^2 + n)$ is divisible as well. So, if $n^3 - n$ is divisible by 3, then $(n+1)^3 - (n+1)$ is divisible as well.



Mathematical Induction

Example 7: Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3 whenever n is a positive integer. [Sp'18,'23]

Solution:

BASIS STEP: For $n=1$, we have $1^3 + 2 \cdot 1 = 3$ is divisible by 3.

INDUCTIVE STEP: Assume that this is true for n and we have to show that it's true for $n+1$.

$$\begin{aligned}\therefore (n+1)^3 + 2(n+1) &= n^3 + 3n^2 + 3n + 1 + 2n + 2 \\ &= n^3 + 2n + 3n^2 + 3n + 3 \\ &= (n^3 + 2n) + 3(n^2 + n + 1) \text{ is divisible by 3;}\end{aligned}$$

$n^3 + 2n$ is divisible by 3 by assumption and $3(n^2 + n + 1)$ is divisible as well. So, if $n^3 + 2n$ is divisible by 3, then $(n+1)^3 + 2(n+1)$ is divisible as well.

Mathematical Induction

Example 8: Prove that for any positive integer n , $\sum_{j=1}^n j(j-1) = \frac{n(n^2-1)}{3}$

Answer :.

Base case: $n = 1$. $\sum_{j=1}^1 j(j-1) = 1 \cdot (1-1) = 0$ and $\frac{n(n^2-1)}{3} = \frac{1(1^2-1)}{3} = 0$, so equality holds when $n = 1$.

Inductive step: Assume that k is a positive integer and $\sum_{j=1}^k j(j-1) = \frac{k(k^2-1)}{3}$.

Show that $\sum_{j=1}^{k+1} j(j-1) = \frac{(k+1)((k+1)^2-1)}{3}$.

$$\begin{aligned}\sum_{j=1}^{k+1} j(j-1) &= \sum_{j=1}^k j(j-1) + (k+1)k \\ &= \frac{k(k^2-1)}{3} + (k+1)k && \text{by the inductive hypothesis} \\ &= \frac{k(k^2-1) + 3(k+1)k}{3} \\ &= \frac{k(k+1)(k-1) + 3(k+1)k}{3} \\ &= \frac{(k+1)[k(k-1) + 3k]}{3} \\ &= \frac{(k+1)[k^2 - k + 3k]}{3} \\ &= \frac{(k+1)[k^2 + 2k]}{3} \\ &= \frac{(k+1)[(k+1)^2 - 1]}{3}\end{aligned}$$

Therefore, $\sum_{j=1}^{k+1} j(j-1) = \frac{(k+1)((k+1)^2-1)}{3}$. ■

Mathematical Induction

Exercise:

1. Use mathematical induction to prove that $3 + 3.5 + 3.5^2 + \dots + 3.5^n = 3(5^{n+1}-1)/4$ whenever n is a nonnegative integer. [Spr'17]
2. Prove that every amount of postage of 12 Taka or more can be formed using just 4 Taka and 5 Taka stamps. [Spr'18]

3.

Prove that

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

Whenever n is nonnegative integer

 [Spr'23]

4. Suppose you have an infinite number of 2 taka and 5 taka notes and you have to pay some amount of money. Use mathematical induction to prove that any amount greater than or equal to 4 taka can be paid with a combination of 2 taka and 5 taka notes. [Aut'23]



Recursion

Recursion refers to a process in which a recursive process repeats itself. Recursive is a kind of function of one and more variables, usually specified by a certain process that produces values of that function by continuously implementing a particular relation to known values of the function.

We use two steps to define a function with the set of nonnegative integers as its domain:

BASIS STEP: Specify the value of the function at zero.

RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.

Such a definition is called a **recursive** or **inductive definition**.



Recursion



Example 1: Suppose that f is defined recursively by $f(0) = 3$, $f(n + 1) = 2f(n) + 3$. Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Solution: From the recursive definition it follows that

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9,$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21,$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45,$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93.$$



Recursion

Example 2: Find $f(1)$, $f(2)$, and $f(3)$ if $f(n)$ is defined recursively by $f(0) = 1$, and for $n=0,1,2,\dots$

i. $f(n+1) = f(n) + 2$

ii. $f(n+1) = 3 f(n)$

iii. $f(n+1) = f(n)^2 + f(n) + 1$ [Aut'17]

Solution:

i. $f(n+1) = f(n) + 2$

From the recursive definition it follows that

$$f(1) = f(0) + 2 = 1 + 2 = 3,$$

$$f(2) = f(1) + 2 = 3 + 2 = 5,$$

$$f(3) = f(2) + 2 = 5 + 2 = 7,$$

ii. $f(n+1) = 3 f(n)$

From the recursive definition it follows that

$$f(1) = 3.f(0) = 3.1 = 3,$$

$$f(2) = 3.f(1) = 3.3 = 9,$$

$$f(3) = 3.f(2) = 3.9 = 27,$$

iii. $f(n+1) = f(n)^2 + f(n) + 1$

From the recursive definition it follows that

$$f(1) = f(0)^2 + f(0) + 1 = 1 + 1 + 1 = 3$$

$$f(2) = f(1)^2 + f(1) + 1 = 9 + 3 + 1 = 13$$

$$f(3) = f(2)^2 + f(2) + 1 = 169 + 13 + 1 = 183$$



Recursion



Exercise:

1. Find $f(1)$, $f(2)$, $f(3)$ and $f(4)$ if $f(n)$ is defined recursively by $f(0) = 3$, $f(n+1) = 2f(n) + 3$ and for $n = 0, 1, 2, \dots$ [Spr'16]
2. Find $f(3)$ and $f(7)$ if $f(n)$ is defined recursively by $f(0) = f(1) = 1$, $f(n+1) = f(n) - f(n-1)$ and for $n \geq 1$ [Spr'17, Aut'23]